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1 RED BALLOON AEROSPACE DEVELOPS INDIA'S FIRST STRATOSPHERIC SUPER-PRESSURE BALLOON

Based in Andhra Pradesh, deep-tech startup Red Balloon Aerospace develops next-generation lighter-than-air platforms for communication, surveillance, and research...



Red Balloon Aerospace - Mission SANA

Founded by Dr Chakravadhanula Venkata Sai Kiran and Sireesh Pallikonda, Red Balloon Aerospace Private Limited was established in May 2025. It develops near-space infrastructure that operates at altitudes of 20-50 kilometres, a region between conventional aircraft and satellites that has remained largely underutilised.

The startup develops advanced lighter-than-air platforms for applications such as communication, surveillance, scientific research, environmental monitoring, and disaster management. Its key products include the HELIX high-altitude airships,

ALTIS tethered aerostat systems, and VISTA super-pressure balloons. These platforms offer satellite-like observation and communication capabilities but operate much closer to Earth, allowing for higher-resolution imaging, lower latency, and significantly greater cost efficiency.

The startup also offers launch services at customer locations and provides extensive customisation options.

RBA launched India's first super-pressure balloon on March 27, 2026 through Mission SANA, marking a historic milestone for the country. In terms of commercialisation, its VISTA super-pressure balloon platform has already completed commercial flights, carrying payloads for commercial aerospace partners during Mission SANA.

ALTIS remains under development, while HELIX, the long-endurance stratospheric airship designed for telecom and cargo applications, is expected to enter commercial deployment during this financial year. According to Dr Kiran, the initial HELIX platform is targeted for completion by December 2026, with flight trials expected to begin in early 2027.

"While drones and aircraft typically operate up to about 10-13 kilometres and satellites function hundreds of kilometres above Earth, there has been little infrastructure in the intermediate near-space

region," explains Dr Kiran. This gap exists largely because the atmospheric conditions at those altitudes are extremely challenging, requiring advanced materials and specialised engineering solutions.

"Red Balloon's proprietary multi-layered nanocomposite materials can withstand harsh temperatures, pressures, and environmental conditions while retaining hydrogen gas for extended periods," claims Dr Kiran. These materials form the basis of the company's lighter-than-air platforms that are protected through provisional patents.

Dr Kiran highlights that the platforms are designed to remain at a stable altitude for prolonged periods, making them suitable for persistent operations. The company's technology enables payloads ranging from 50 kilograms to 20 tonnes, depending on mission requirements. Customers can choose from rideshare missions, where multiple payloads share a launch, or dedicated launches tailored to specific needs.

Dr Kiran shares that nearly all aspects of design, development, manufacturing, and production are handled in-house. Even many of the manufacturing machines used in production have been designed and built by the company because suitable equipment was not readily available on the market.

"Around 80-90 per cent of the